

Signed numbers: addition and subtraction

Initial assessment • Work out all four questions. Show your working.

1. $-2 + 9$

2. $6 - 11$

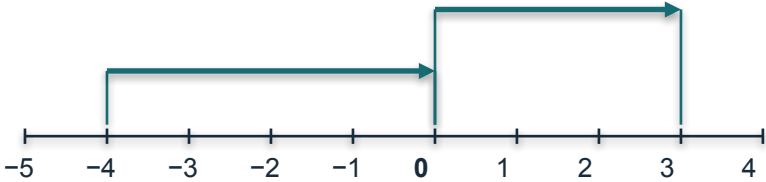
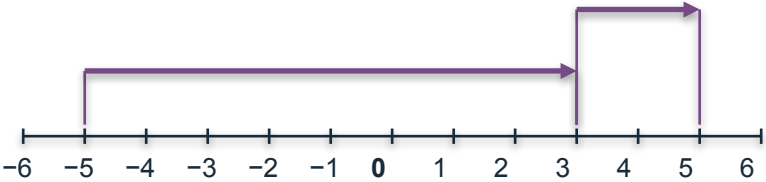
3. $-7 - (-10)$

4. $5 + (-9) - (-6)$

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Worked demo • Follow each calculation.

Adding a negative moves left: $4 + (-7) = 4 - 7 = -3$.
Subtracting a negative moves right: $6 - (-3) = 6 + 3 = 9$.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	$-4 + 7$	$6 - (-3)$	$-5 + 8 - (-2)$
Read the Signs	Start at -4 ; add positive 7 .	Start at 6 ; subtract negative 3 .	Add positive 8 , then subtract negative 2 .
Rewrite the Calculation	$-4 + 7$ is already an addition.	$6 - (-3) = 6 + 3$.	$-5 + 8 + 2$.
Calculate	Move 4 right to 0, then 3 right: 3. +4 then +3: total +7 	$6 + 3 = 9$. subtract -3 = add 3 	$-5 + 8 = 3$; $3 + 2 = 5$. +8, then +2 
Check the Answer	$3 - 7 = -4$. Answer: 3 .	$9 + (-3) = 6$. Answer: 9 .	$5 + (-2) = 3$; $3 - 8 = -5$. Answer: 5 .

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Step-by-step practice 1 • Show your working.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	$-6 + 9$	$4 + (-7)$	$-8 + 5 - (-4)$
Read the Signs	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.
Rewrite the Calculation	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.
Calculate	Work from left to right. Show each result.	Work from left to right. Show each result.	Work from left to right. Show each result.
Check the Answer	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.

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Step-by-step practice 2 • Show your working.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	$3 - 8$	$-5 - (-9)$	$7 - (-2) + (-12)$
Read the Signs	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.
Rewrite the Calculation	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.
Calculate	Work from left to right. Show each result.	Work from left to right. Show each result.	Work from left to right. Show each result.
Check the Answer	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.

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Independent practice 1 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-3 + 8$	1. $5 - (-2)$	1. $-4 + 7 - (-3)$
2. $2 - 7$	2. $7 + (-9)$	2. $6 - 9 + 2$
3. $-6 + 4$	3. $-4 + (-6)$	3. $-5 + (-3) + 10$
4. $-2 - 5$	4. $-8 - (-3)$	4. $8 - (-2) - 13$
5. $-9 + 9$	5. $-5 - (-5)$	5. $-7 - (-4) + 1$
6. $4 - 10$	6. $0 + (-7)$	6. $3 + (-8) - (-5)$

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Independent practice 2 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-4 - 6$	1. $9 - (-4)$	1. $-6 + 9 + (-4)$
2. $-11 + 11$	2. $-6 + (-7)$	2. $4 - (-6) + (-10)$
3. $6 - 15$	3. $0 - (-9)$	3. $7 - 10 - (-5)$
4. $-7 + 10$	4. $3 + (-8)$	4. $-8 - (-2) - 1$
5. $-8 + 3$	5. $-3 - (-8)$	5. $-9 + 5 - (-7)$
6. $5 - 12$	6. $-10 - (-6)$	6. $5 + (-12) + 3$

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Independent practice 3 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-8 + 12$	1. $-9 + (-2)$	1. $10 - 14 + (-2)$
2. $-5 - 4$	2. $8 - (-5)$	2. $6 + (-9) - (-3)$
3. $-10 + 6$	3. $-7 - (-4)$	3. $-2 - (-8) + 5$
4. $9 - 14$	4. $6 + (-11)$	4. $-3 + (-6) - (-4)$
5. $7 - 16$	5. $-4 - (-10)$	5. $9 - (-3) + (-7)$
6. $-12 + 12$	6. $-12 - (-7)$	6. $-11 + 8 - 2$

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Independent practice 4 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-12 + 7$	1. $0 - (-6)$	1. $-12 + 7 + 8$
2. $11 - 18$	2. $-11 - (-8)$	2. $4 - (-5) + (-12)$
3. $-6 - 8$	3. $5 + (-13)$	3. $-6 - (-9) + (-3)$
4. $-9 + 14$	4. $-7 - (-12)$	4. $5 - 13 - (-2)$
5. $-15 + 15$	5. $12 - (-3)$	5. $8 + (-11) - 4$
6. $8 - 13$	6. $-4 + (-9)$	6. $-10 + 6 - (-7)$